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Challenges and opportunities for social sustainability in education: a scoping review

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ABSTRACT

Social sustainability in education is crucial in fulfilling Agenda 2030, yet an area not well represented in research. This scoping review aims at investigating this gap, covering the years 2018 to 2023, with a focus on how social sustainability is defined and implemented in education and what challenges and opportunities for achieving social sustainability that have been identified in the education research in Web of Science, ERIC, and Scopus databases. A total of 120 articles were identified and screened, 76 as full text, and out of these 42 were included in the final analysis. While all articles highlight the vital role of teaching social sustainability in education, some also point out that its broader implementation is hindered by a lack of theoretical and conceptual clarity. This vagueness also creates difficulties in identifying and assessing the impact of social sustainability issues in education. This review focuses on how social sustainability has been defined and implemented in education, what challenges and opportunities for social sustainability issues in education that can be identified, and furthermore how education can contribute to overall sustainability within the frameworks of Education for Sustainable Development (ESD) and Education for Sustainable Development Goals (ESDG).

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Introduction

In the face of unprecedented global challenges—ranging from social inequalities to environmental crises—the urgency of sustainable development has never been greater (IPBES, 2019; IPCC, 2023). These crises not only destabilize ecosystems and the essential services they provide but also pose significant threats to global food security, public health, and socio-economic stability. Moreover, they intensify existing inequalities and increase vulnerabilities, particularly among marginalized communities (Zagama et al., 2023).

The increasing focus on sustainability marks a significant shift in how we understand the relationship between humans and nature, and between people, and reflects an effort to integrate rising concerns about environmental challenges with socio-economic issues (Hopwood et al., 2005). It also indicates that sustainability is not only an environmental issue, but has a social dimension (Dillard et al., 2009; Griessler & Littig, 2005; Magis & Shinn, 2009).

Social sustainability is usually, but not always, referring to United Nation's Agenda 2030 and its Sustainable Development Goals (SDGs), which were introduced in 2015. The Agenda 2030 is a global plan of action for people, planet, prosperity, peace and partnership. In its declaration, it is stated that it is necessary to take bold and transformative steps to shift the world onto a sustainable and resilient path. All countries and all stakeholders, collaborating as partners, are involved in its implementation. Agenda 2030 is organised into 17 Sustainable Development Goals (SDGs) and 169 targets that seek to build on the Millennium Development Goals and complete what these did not achieve. It is emphasised that the SDGs are integrated and indivisible and balance the three dimensions of sustainable

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development: the economic, social and environmental ‘pillars.’ The SDGs aspire to stimulate action in areas of critical importance for humanity and the planet to achieve sustainable development goals in 2030 (UN, 2015) and ‘constitute the most influential framework for the global development agenda at the present time’ (Van Tulder et al., 2021, p. 1).

In comparison with environmental and economic dimensions of sustainability, social sustainability is a relatively new concept in policy discourse and academic literature (Isgren & Longo, 2024, p. 820) and has received significantly less attention. However, while environmental and economic sustainability often dominate public discourses, social sustainability is just as crucial. The importance of the social dimension of sustainability was emphasised already in the Brundtland (1987) Report ‘Our Common Future’, although the report has also been criticised for holding a vagueness of the specifics of the social dimension and how it was to be understood (Borowy, 2021). In the more recent Agenda 2030 and its 17 SDGs, the social dimension of sustainability is more concrete and operationalizable and refers to specific goals such as eradicating poverty and achieving gender equality (Langergaard & Krøjer, 2023, p. 4–5).

The term social sustainability encompasses a wide range of meanings and perspectives and refers to matters such as justice, inclusion, equity, equality, democracy, and social capital, as well as human skills. Social sustainability issues emphasize the creation of inclusive, equitable, and cohesive societies where everyone can thrive. In the realm of education, social sustainability highlights the role of schools in promoting diversity, equity, and the well-being of both students and communities. Social sustainability in education refers to the cultivation of inclusive learning and the shaping of values, attitudes and behaviours of learners to become engaged and responsible citizens contributing positively to society. It also addresses issues like human rights, and safeguarding social values, cultural identity, and traditions (Ajmal et al., 2018; Frank et al., 2024; Vallance et al., 2011).

As societies grapple with challenges like growing inequality and social fragmentation, the role of education in promoting social sustainability is increasingly important. Schools not only prepare students for the workforce but also shape their sense of responsibility towards their communities, making education a cornerstone for sustainable social development (UNESCO, 2017).

Although education is explicitly formulated as a stand-alone social sustainability goal—SDG 4—targets related to education are also contained within other SDGs (Table 1), making education not only an integral part of sustainable development, but according to UNESCO also its key enabler (UNESCO, 2017).

Target 4.7 under SDG 4, which aims to ensure inclusive and equitable quality education and to promote lifelong learning opportunities for all, states that by 2030, we should ensure that all learners acquire

Table 1. SDGs and targets related to social sustainability.

SDG 1: No Poverty
Targets include eradicating extreme poverty, implementing social protection systems, ensuring equal rights to economic resources, and building resilience to climate-related events and economic shocks
SDG 2: Zero Hunger
Targets include ending hunger, achieving food security, improving nutrition, and promoting sustainable agriculture, with special attention to small-scale food producers and equal access to land and resources.
SDG 3: Good Health and Well-being
Targets include reducing maternal mortality, ending epidemics of communicable diseases, ensuring universal health coverage, and promoting mental health and well-being.
SDG 4: Quality Education
Targets include ensuring inclusive and equitable quality education, promoting lifelong learning opportunities, and achieving gender equality in education.
SDG 5: Gender Equality
Targets include ending discrimination and violence against women and girls, recognizing unpaid care and domestic work, and ensuring women's full and effective participation in leadership.
SDG 8: Decent Work and Economic Growth
Targets include promoting sustainable economic growth, achieving full and productive employment, and ensuring decent work for all, including youth and persons with disabilities.
SDG 10: Reduced Inequalities
Targets include reducing income inequality, promoting social, economic, and political inclusion, ensuring equal opportunities, and facilitating safe migration and mobility.
SDG 11: Sustainable Cities and Communities
Targets include ensuring access to adequate, safe, and affordable housing and transportation, enhancing inclusive urbanization, and providing access to green and public spaces.
SDG 16: Peace, Justice, and Strong Institutions
Targets include promoting peaceful and inclusive societies, providing access to justice for all, reducing violence, and ensuring accountable and inclusive institutions.

knowledge and skills needed to promote sustainable development (UNESCO, 2017). In this, Agenda 2030 clearly reflects the importance of an appropriate educational response to the call for sustainability.

The launch of Agenda 2030 in 2015 was supported by the UNESCO's Global Action Programme (GAP), which was a follow-up from the UN Decade of Education for Sustainable Development (2005–2014) that had envisioned a world 'where everybody has the opportunity to benefit from education and learn the values, behaviour and lifestyles required for a sustainable future and for positive societal transformation' (UNESCO, 2018). In 2020, GAP was replaced by the Education for Sustainable Development for 2030 Roadmap which places 'a stronger focus on education's central contribution to the achievement of the Sustainable Development Goals' (UNESCO, 2020). Other global initiatives include the Mission 4.7 which specifically targets SDG 4.7 and focuses on integrating sustainability principles into education systems worldwide, fostering global citizenship and promoting a culture of peace and non-violence, and the SDG Academy, which offers flexible online courses and open-access educational resources focusing on the 17 SDGs.

Since the introduction of Agenda 2030, critics have noted 'a tendency of educational institutions to embrace, uncritically, education for sustainability and the SDGs' (Adelman, 2018), leading to the development of an Education for Sustainable Development Goals (ESDG) which focuses specifically on meeting the sustainability targets under the SDGs and 'unreflexively aims to encourage learners to "take action for sustainable development"' (Kopnina, 2020b, p. 282; UNESCO, 2017). By contrast, Education for Sustainable Development (ESD) addresses some of the criticism that has been placed on (industrial and economic) 'development' before 'sustainability' (of a finite planet) and the incompatibility of several SDGs (Kopnina, 2020a, 2020b; Washington, 2015), including the assumption that 'what is good for society is good for 'nature' or vice versa' (Isgren & Longo, 2024, p. 824). ESD seeks 'to foster learners to be creative and responsible global citizens who critically reflect on the ideas of sustainable development and the values that underlie them' (Kopnina, 2020b, p. 282; Spannring, 2019). According to ESD approaches, the integration of sustainability values into all areas of learning can foster a rethinking of current practices and promote behavioural changes towards a more sustainable society (Franco et al., 2019; Yarime et al., 2012).

Education is recognized as a key driver in achieving sustainable development goals (Buerkle et al., 2023, p. 2), but this presupposes the desirability of sustainable development as it is defined by Agenda 2030 and its SDGs, a notion that is not universally agreed upon (Kopnina, 2020b, p. 281), and which includes criticism against sustainable development and its many policy manifestations for its alleged Western bias, growth-centrism, and anthropocentrism (Banerjee, 2003; Demaria et al., 2023).

Despite efforts to achieve the SDGs (ESDG) or address sustainability issues more broadly in education (ESD), social sustainability in education remains neither clearly defined nor effectively implemented. Ten years after the inception of the 2030 Agenda, this ambitious action plan has encountered significant obstacles in translating its aspirations into tangible actions. Although the global goals have inspired discourse and contributed to normative and institutional adjustments, they have not realised their potential as change agents and have remained insufficient to deliver on promises of global sustainability, coherence, integration, and equality, especially when it comes to developing countries and those 'left behind' (Biermann et al., 2022). Overall, the improvements under the adoption of the SDGs have been slow and insufficient (Farias et al., 2020; Klingenberg & Rothberg, 2022) and there has been a notable delay in the field of education (Ruge et al., 2022). It has been argued that if we continue with the same efforts as before, we will not reach the sustainability goals by 2030, and not even by 2050 (Randers et al., 2018).

When it comes to exploring social sustainability, one obstacle has been that the concept 'lacks a coherent, clear and utilizable definition' (Eizenberg & Jabareen, 2017, p. 2), and instead there is a wide array of definitions with none of them being universally accepted (Langergaard & Krøjer, 2023). Because of the intricate complexity and diversity inherent in social sustainability, it has been described as a 'mess of diversity' (Kenter et al., 2019), and 'a mysterious and somewhat fuzzy concept,' that is 'undertheorized,' 'in chaos' (Boström, 2012; Eizenberg & Jabareen, 2017; Vallance et al., 2011; Woodcraft, 2012, p. 29) and 'characterized by wickedness since social norms, values, ethics, and the understanding of good and bad are as diverse as the global population' (Šilenskytė et al., 2024, p. 1). Social sustainability is sometimes presented as the sustainability of sociality itself, such as sustainability of community (Krøjer, 2020), and

sometimes as the condition for achieving green goals through collective action (Langergaard, 2020), for instance in urban planning.

Definitions of social sustainability often include social equity and access to important services, social inclusion and cohesion for all people and communities (Dempsey, 2009; Murphy, 2012), but social sustainability is also sensitive to the context of its application (Janssen & Basta, 2024; Sze et al., 2018). What social sustainability offers, it seems, is an 'catch-all term' for different social issues like equity, equality, justice, well-being, basic needs, democracy, participation, empowerment, employment, education, security and social capital (Isgren & Longo, 2024, p. 821), but there is little consistency in the combinations or emphasis placed on these concepts, and social sustainability seems to operate as little more than a container for whatever is in-vogue in social policy discourses (Davidson, 2010).

Consequently, it has been easier to focus on the economic and environmental sustainability issues (Boyer et al., 2016) and the social dimensions of sustainability have received little empirical and theoretical investigation (Ajmal et al., 2018; Eizenberg & Jabareen, 2017; Larimian & Sadeghi, 2021). Despite numerous contributions and extensive work, 'a universally accepted definition of 'social sustainability' remains elusive, indicating that substantial conceptual work is still required' (de Fine Licht & Folland, 2025, p. 46). In contrast, the vagueness and openness of the concept of 'sustainability' is for some its strength and usefulness as a 'boundary object' (Scoones, 2007), making the concept of sustainability useful as an analytical tool 'despite (and in part thanks to) this ambiguity' (Isgren & Longo, 2024, p. 822).

Another problem with social sustainability is its assessment, as reporting systems and numerical data fail to capture social evaluations, stakeholder perspectives, and the complex dynamics of their interests, relationships and emotions (Šilenskytė et al., 2024, p. 5). It has therefore been suggested that social sustainability assessments need to move away from relying solely on 'objective data' and 'scientific evidence' to drive changes in human perspectives and behaviour. Achieving progress towards social sustainability requires combining rigorously measured facts with practitioner insights and societal interpretations (Vallance et al., 2011, p. 347). Moreover, social sustainability and quality education must be developed through cross-disciplinary approaches, requiring broad and diverse perspectives (Azmat et al., 2023).

Nevertheless, Agenda 2030 and its targets for sustainable development under the SDGs presents persuasive arguments and many education institutions have explored ways to implement sustainability issues in education (Leal Filho et al., 2021). The ESDG approaches generally fall into two categories: teaching SDGs directly and applying SDGs as a pedagogical tool. The first approach, teaching SDGs, focuses on equipping learners with the knowledge and skills to promote sustainable development, including through education on human rights, gender equality, peace, global citizenship, and cultural diversity (Buerkle et al., 2023, p. 2). The second approach emphasizes embedding sustainable development principles into the teaching process itself, rather than just using SDGs as educational content. This involves enhancing teaching quality by equipping educators with sustainability competencies (Cebrián et al., 2020; Nogueiro & Saraiva, 2023), which encompass cognitive skills, practical abilities, and ethical values (Cebrián & Junyent, 2015).

However, developing pedagogical strategies for education, that integrate sustainable development and its related challenges, is widely considered a complex endeavour (Bascopé et al., 2019; Leal Filho et al., 2015). A major barrier to developing and implementing solutions to the current crisis lies in the complexity of sustainability-related issues, often classified as 'wicked problems,' characterized by high societal risks but difficult to resolve due to incomplete or conflicting information, the involvement of numerous stakeholders with diverse perspectives, significant economic investments, and their interconnectedness with other challenges (Frank et al., 2024, p. 1478).

To tackle the highly complex sustainability challenges, the development of critical consciousness has been suggested as a pedagogical approach within ESD that may empower students to engage in critical reflection, thought and action towards social justice by examining existing power structures, committing to addressing injustices and making collaborative efforts to dismantle oppressive systems. Educators must involve knowledge and values that are focused on social justice issues, such as democracy, diversity, equity and human rights, and by engaging in critical consciousness, students can better understand how various social, political, cultural, economic, and ecological systems contribute to unsustainability,

inequity, and injustice (Bosio & Waghid, 2019). The decolonisation of education curricula also requires educators to critically reflect on existing privileges in education, historicize and explore identity issues and how these may be embedded in education, and engage in decolonial reflexivity to encourage stronger engagement with issues of race and privileges (Allen & Girei, 2023, p. 1).

While both ESD and ESDG advocate for integrating sustainability into education, ESD has a broader focus on sustainability principles across various educational contexts, whereas ESDG specifically emphasizes the alignment of educational efforts with the Sustainable Development Goals, stating that its focus is on becoming a 'universalizing discourse.' However, according to critiques, unreflective acceptance of the SDGs as a universal good by schools and universities is a cause of concern (Engelman, 2013) and may lead to the development of an 'ignorant society,' in which citizens accept prefabricated and clichéd viewpoints (Brey et al., 2009, p. 26).

To promote critical reflection in schools, Schmitz et al. (2024, p. 431–432) present seven strategies that educators may use to foster 'sustainability mentality' among learners: integrating sustainability into the curriculum, using practical examples, advocating for systems thinking (facilitate learners' comprehension of intricate global phenomena), promoting emotional and spiritual intelligence (delve into their personal emotions and ideals), modelling sustainable behaviours (educators as role models), promoting collaborative learning, and promoting continuous learning. Cox-Petersen (2011) has also suggested an increased dialogue and partnership between educational institutions and disciplines to foster sustainability mentality among learners, by engaging in interdisciplinary collaborations, joint programs, research partnerships, guest lectures and workshops, and community projects. Another learning tool is the 'Inner Development Goals,' which assumes that sustainable development also needs a change of mindset at the individual level (Inner Development Goals, 2021). By applying these education tools and models and using the diversity humans have to offer, learners can build imaginative and more sustainable worlds and translate them to real-world action (Zaidi, 2019).

Previous research about social sustainability issues in education have identified a number of challenges for its implementation, including a risk of failing to meet SDG targets, a lack of critical reflection around the premises for 'sustainable development,' a lack of attention to social dimensions of sustainability, difficulties with agreeing on a definition of 'social sustainability' to make it operational, subsequent difficulties with assessing progress, and 'wicked problems' that stand in the way of realising complex and sometimes conflicting SDG targets.

Despite, or perhaps because of these problems, and given the importance placed on education as a driver of sustainability, this review investigates how social sustainability issues have been researched within the education field between 2018 and 2023 and how social sustainability is defined and interpreted across disciplines and levels of education. It also investigates what different challenges and opportunities for social sustainability that have been identified in the research and how, according to the research, social sustainability can be achieved through education and meet the SDG targets and sustainability goals before 2030.

Rational and aim

With merely five years to go before the Agenda 2030 and its 17 SDG targets are to be achieved, and given the importance placed on social sustainability to achieve overall sustainability, especially through education, it is prudent to investigate how social sustainability goals and targets are currently addressed and implemented at various levels of education. To this end, I carried out a scoping review to investigate how social sustainability is defined in education and education research, at what levels of education and where the research is carried out and what challenges and opportunities in the implementation of social sustainability issues that have been identified.

The search was not limited to any specific SDG, as it was of interest to find out how SDG 4, which is most obviously related to social sustainability and education, is perhaps related to other SDGs in the research. To contribute in a meaningful way, the review focused on understanding trends and patterns in the current body of literature around social sustainability in education and the research was therefore limited to cover the years 2018 to 2023. The analysis highlighted key issues that can contribute to a better understanding of challenges and opportunities for social sustainability in education, and how it

relates to ESD and ESG approaches. It was also of interest to explore at what levels of education that research into social sustainability was carried out, in what disciplines, and in what locations.

The following research questions have guided this scoping review:

1. How is social sustainability defined and interpreted within education?
2. What opportunities and challenges for social sustainability in education can be identified in the research? How is social sustainability related to ESD and ESG approaches in education?

Methods and materials

To identify what social sustainability issues that occur in education research, a scoping review can map the literature and provide useful overviews of research within this specific field, identify central concepts and knowledge gaps, facilitate the development of theory, and identify new areas of research. These kinds of reviews are particularly useful in understudied bodies of literature about large and complex issues (Peters et al., 2015). In these contexts, scoping reviews offer a rigorous and transparent research method (Arksey & O'Malley, 2005) by investigating a specific topic, using well-defined inclusion and exclusion criteria, and applying clearly defined research procedures (Peterson et al., 2017).

This review has used a framework for scoping reviews including a process in five stages: (1) the identification of research questions, (2) the identification of relevant studies, (3) selection of studies, (4) data charting, and (5) collation, summarizing and reporting results (Arksey & O'Malley, 2005). In this review, stages (2–3) involved two team members with expert knowledge within the chosen field of research, who independently carried out blind assessments of initial search results (see Daudt et al., 2013 for a similar approach). The final inclusion of publications for the review is the result of a joint agreement over the relevance of the publications finally selected for the study.

The nomenclature of the chosen field should also be considered to avoid unnecessary limitations of the literature search and the research (Karrow et al., 2022). Because the purpose of this study is to carry out a broad investigation about social sustainability research in education between the years 2018 and 2023, the keywords selected were therefore open enough to identify research across levels of education, disciplines and locations of research.

On September 13, 2023, a comprehensive search of peer-reviewed journals and publications was carried out based on keywords that included social sustainability (sustainab*, 'social sustainab*'), education (educat*), programs (program*), and courses (cours*) in titles, abstracts, or keywords. Boolean operators (And/Or) were used to ensure that the search yielded maximum results, and active filters were used to limit the search. The search, which focused on the following three databases, yielded 120 articles published over a six-year period from 2018 to 2023: ERIC, Scopus (Elsevier), and Web of Science (see the search strategy protocol in Table 1).

The three databases were selected based on their relevance for the purpose of this scoping review, i.e. investigating research about social sustainability in education.

The following inclusion criteria were applied in the search:

Inclusion criteria:

1. Recent research—publication date within the last six years (2018–2023).
2. Publication type: journal articles, books/book chapters and Ph D theses.
3. Peer reviewed publications.
4. Addressing social sustainability issues in education.

The following exclusion criteria were applied in the research:

1. Research published before 2018 or after 2023.
2. Publication types that did not constitute journal articles, books/book chapters and Ph D theses.
3. Non-peer reviewed publications.
4. Addressing topics irrelevant to social sustainability issues in education.

Duplicates and articles not written in English or Swedish were excluded.

A total of 197 records were retrieved, and after removing 77 duplicates, 120 records remained (Figure 1) and were transferred to Rayyan for Title and Abstract Screening.

The author of this study invited two colleagues with expertise within the chosen field of research to carry out blind assessments of the 120 records in Rayyan, reviewing keywords and the content of

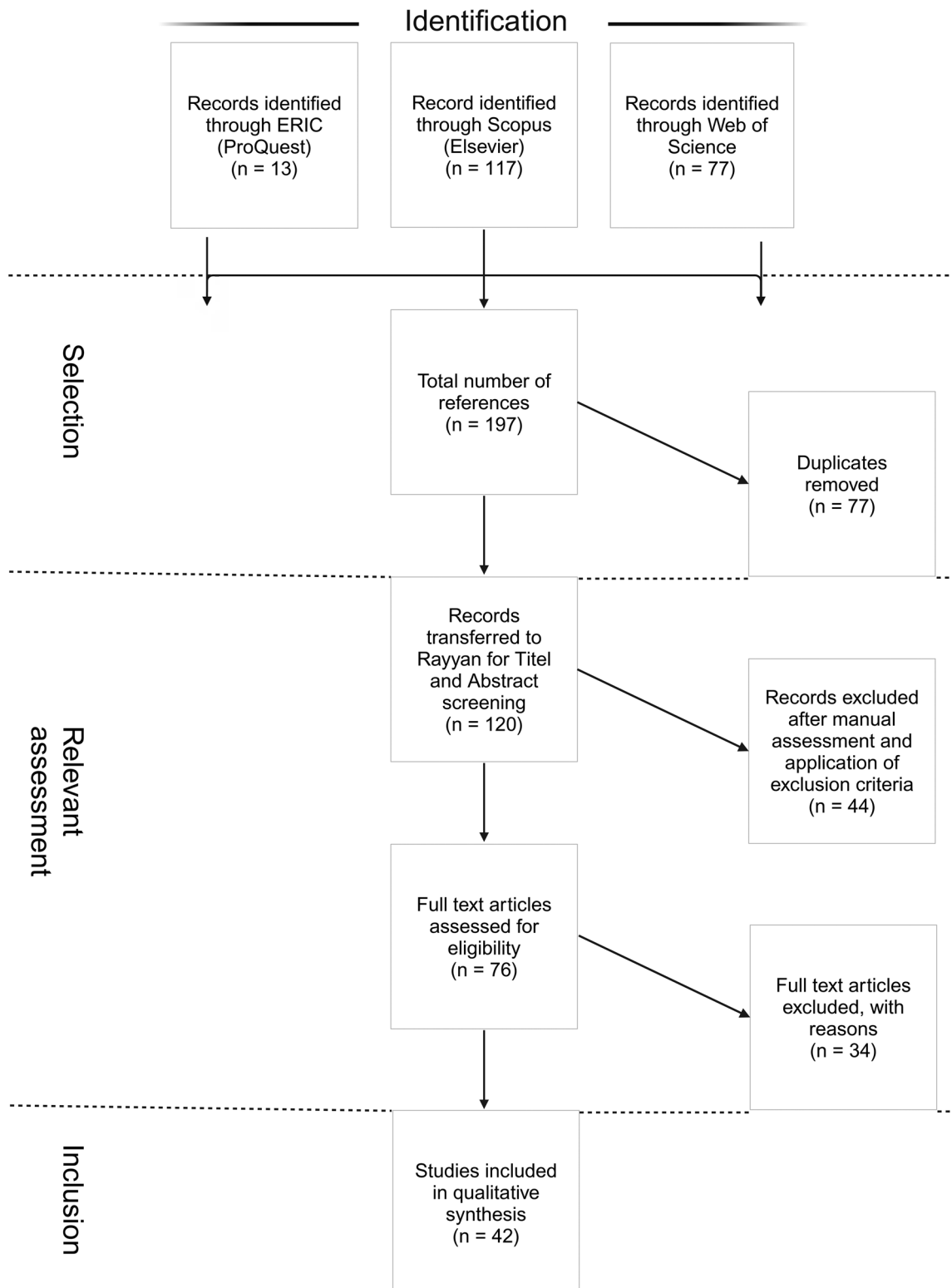


Figure 1. Search strategy protocol.

abstracts using the inclusion criteria. The assessment was carried out in two stages; first with one of the colleagues, followed by consultation over resulting mismatches, and then with the other colleague, assessing the synchronised list from stage 1, followed by consultation over remaining mismatches to reach a final list of 76 records for further assessment of full texts.

The list of records was transferred to EndNote to facilitate the import and review of full text articles by the first author of this study. One further duplicate that was not detected in the initial database search, was removed from the list of records. Five records were inaccessible in full text due to paywalls, two were inaccessible because they were only published as abstracts and two were published as conference proceedings and were subsequently removed. 24 records were deemed, after reviewing the full texts using the exclusion criteria, i.e. did not include definitions of social sustainability in the full text versions and/or did not apply to education, as non-applicable to the study and were therefore removed. A list of 42 relevant records were selected for the final analysis (Table 2).

Data charting

All the 42 articles selected for the review were read in full. A spreadsheet was created to chart relevant data, guided by the research questions. Columns in the spreadsheet included information about definitions or interpretation of and association with 'social sustainability,' and challenges and opportunities to social sustainability in education. Definitions and interpretations of and association with social sustainability were charted whenever the articles explicitly stated their own definitions, using other's definitions and/or whenever the author/s explained how education issues were associated with social sustainability. Challenges were charted as obstacles or potential obstacles for social sustainability in education whenever the author/s mentioned these in the texts, and opportunities were charted according to explicit mentioning of what possibilities that the author/s identified with social sustainability issues in education.

Collation, summarizing, and reporting results

The data charting sheet was used to organize information into categories guided by the research questions. To answer research question 1, all the different concepts that were used to define, interpret and/or associate with 'social sustainability' in each study were documented and analysed. To answer research questions under 2 stated challenges and opportunities for social sustainability and different approaches to how to achieve social sustainability in education were documented and analysed.

Results

Research question 1: How is social sustainability defined and interpreted within education?

Screening the 42 articles for the kinds and frequency of words used to define and describe social sustainability in education has resulted in a list of over 80 different words and concepts used in the articles (Table 3).

All articles in this review are included based on mentioning 'social sustainability' in titles, keywords and/or abstracts, but depending on the topic of the specific article in this review, references to social sustainability issues occur in different contexts. In some of the articles, social sustainability is explained using several defining concepts, for example that '(s)ocial sustainability is a broad term that includes the preservation and development of stable societies with social justice, equal rights, citizenship, participation, well-being, health, education, and safety for all people in the community' (Bergan et al., 2021, p. 4268). However, some observations can be made regarding the distribution of articles along disciplines, education levels and location of studies. Articles representing social and natural and applied sciences dominate this review numerically, whereas the number of articles focusing on business and humanities are relatively few (Table 4).

Articles focusing on higher education number 25 out of 42 in this review (Table 5) and it can be noted that 12 out of the 16 articles focusing on natural and applied sciences also focus on higher education (Tables 4 and 5).

Table 2. Articles included for final analysis.

No	Author/s and year of publication	Discipline	Level of education	Country/ location of study	Focus of article
1.	B. Balakrishnan et al., 2021	Natural and applied sciences	Higher education	Malaysia and Japan	Examines the impacts of education for sustainable development subject(s) offered at universities
2.	V. Bergan et al., 2021	Social sciences	Early age/ Preschool	Norway	Explores how 'community of practice' can be recognised in foraging and garden projects
3.	M. Biasutti et al., 2019	Social sciences	Elementary	Italy	Assesses a professional development training course in intercultural education
4.	S. Boldermo, 2019	Social sciences	Early age/ Preschool	Norway	Investigates the relationship between early childhood education for social sustainability and migrant children's negotiations of social identity and belonging in early childhood
5.	S. Boldermo & Ødegaard, 2019	Social sciences	Early age/ Preschool	N/A	A review of research articles about education for sustainability, primarily in early childhood
6.	F. Borg & Gericke, 2021	Social sciences	Early age/ Preschool	Sweden	The study aims to increase knowledge about preschool teachers' teaching practices relating to social sustainability, using Education for Sustainability as a conceptual framework
7.	K. Breznik et al., 2021	Natural and applied sciences	Higher education	Slovenia	Dimensions of sustainability in mission statements from engineering institutions are compared with other higher educational institutions
8.	S. Ceylan & Soygeniş, 2019	Natural and applied sciences	Higher education	Türkiye	Examines the social aspects of sustainability and its relation to architecture, with respect to the strong connection between the society and the built environment
9.	T. B. M. Chowdhury et al., 2020	Natural and applied sciences	N/A	N/A	Seeks to identify the role of science education in promoting an active, scientifically literate, citizenry ready to address sustainable development goals
10.	N. Cuenca-Soto et al., 2023	Natural and applied sciences	Higher education	Spain	Aims to explore the effects of a critical (eco) feminist approach implemented through a critical feminist service-learning program
11.	M. Domingues, 2021	Social sciences	Early age/ Preschool	U.S.A.	Explores the lessons learned from using a gender-fluid persona doll in early childhood education for social sustainability
12.	E. Erjavec, 2020	General	Higher education	EU	Extends the analysis of the contribution of education as a public intangible capital to the economic growth
13.	O. Franck & Osbeck, 2018	Social sciences	Elementary	Sweden	Aims to present some prerequisites for the development of narrative methods that focus on the vision of a good society without compromising individual integrity
14.	K. A. A. Gamage et al., 2022	General	Higher education	N/A	Offers insights into incorporating the sustainability-happiness nexus into sustainable universities to enhance the social dimension of sustainability
15.	S. Guo et al., 2020	Humanities	N/A	Thailand	Examines the language landscape in Chiangmai, Thailand, and the linguistic beliefs of local Thai Chinese language learners
16.	H. Gürgen Şimşek & Ö. Erkin, 2022	Natural and applied sciences	Higher education	Türkiye	Aims to determine sustainable development awareness and related factors in nursing students
17.	P. Hallinger et al., 2020	Natural and applied sciences	N/A	N/A	Using a bibliometric review to address the research on simulations and serious games used in educating for sustainability
18.	S. H. M. Harizan & S. A. M. Shah, 2022	Social sciences	Lifelong learning	Malaysia	Considers the question of 'How does online and distance education relate to sustainability?'
19.	M. Heldal et al., 2021	Social sciences	Early age/ Preschool	Greece	Discusses how an Early Childhood Education and Care (ECEC) institution in a refugee camp can promote social sustainable education
20.	M. Hoveskog et al., 2018	Business	Higher education	Sweden	Describes an Experiential Workshop for university undergraduates in which the Service-Learning pedagogic approach is taken and Flourishing Business Canvas is applied as a tool for collaborative visual business modelling
21.	B. P. Hung, 2023	General	Higher education	Vietnam	Aims to investigate the effects of urban institutions of higher education on social sustainability of the community
22.	I. Ibrahim, 2020	Natural and applied sciences	Higher education	United Arab Emirates	Explores the social potentials in the courtyards as an architectural element used to recognize the traditional architecture in Arab countries

(Continued)

Table 2. Continued.

No	Author/s and year of publication	Discipline	Level of education	Country/ location of study	Focus of article
23.	K. I. Kazamia & Kafaridou, 2019	Natural and applied sciences	Higher education	Cyprus	Focuses on how to enforce the social pillar of sustainability through design education
24.	N. E. Kordi et al., 2022	Natural and applied sciences	Higher education	Malaysia	Explores the incorporation of the social dimension in the Civil Engineering curricula of an undergraduate program
25.	M. Larkins et al., 2018	Social sciences	Higher education	N/A	Aims to examine the textual coverage of the topic of public engagement in leading English language sustainability textbooks
26.	T. Liu et al., 2022	General	Higher education	China	Aims to examine the relationship between regional sustainability and higher education in China using fixed-effects panel modelling
27.	H. Liwång, 2020	Natural and applied sciences	Higher education	Sweden	Aims to identify and document the pedagogic lessons for a course module where sustainable development is discipline specific
28.	L. Lo Presti et al., 2023	Business	Higher education	Europé	Investigate the commitment of universities in terms of inclusion and social justice in the light of the opportunities offered by university public engagement
29.	R. Luchoro-Parrilla et al., 2021	Social sciences	N/A	Spain	Studies from an ethnomotor perspective the different characteristics of two different groups of Traditional Sporting Games in the Canary Islands
30.	G. Malandrakis et al., 2019	Social sciences	Higher education	Greece	Explores student-teachers' progress in understanding social urban sustainability and how it is affected through the development of digital stories and engagement in civic actions
31.	L. Marcelino et al., 2019	Natural and applied sciences	N/A	N/A	Studies renowned papers by Green Chemistry pioneers and reviews on the field regarding how they address the social dimension of sustainability
32.	E. E. Odegaard, 2021	Social sciences	Early age/ Preschool	Norway	Aims to identify the components and features of a signature pedagogy for sustainability in early childhood education and care to respond to the call for tradition and innovation in early childhood education
33.	C. T. Oropilla & Ødegaard, 2021	Social sciences	Early age/ Preschool	N/A	Explores the inclusion of intentional intergenerational engagements and programmes in kindergartens as approaches towards sustainable futures for children
34.	A. Pasha & Sreekanth, 2022	Natural and applied sciences	Higher education	India	Discusses how preparing students, staff, whole campus, and community to be more proficient decision makers of future environment can be achieved by integrating sustainability into educational institutions
35.	I. S. Rampasso et al., 2023	Natural and applied sciences	Higher education	Brazil	Aims to present the main initiatives developed in the discipline to promote sustainability in engineering students
36.	M. Rey-García & V. Mato-Santiso, 2020	Business	Higher education	Spain	Aims to understand the roles that social capital and real-world learning may play in enhancing the effects of university education for sustainable development on social sustainability
37.	A. R. Sadownik and J. Gabi, 2021	Social sciences	N/A	N/A	Presents some implications that using post-human language (to understand belonging) may have for policy, Early Childhood Education and care practice and theory
38.	T. Siller et al., 2022	Natural and applied sciences	Higher education	U.S.A.	Argues that engineering curricula ought to include sustainability and social justice so future engineers are trained to understand both societal and technical implications of their work
39.	S. Simmonds & Ajani, 2022	Social sciences	Higher education	South Africa	Presents data that emanated from qualitative interviews conducted with postgraduate teacher education students to unlock their concerns and aspirations for a decolonised curriculum in higher education
40.	S. A. Syed Zakaria et al., 2020	Natural and applied sciences	Higher education	Malaysia	Presents results pertaining to the exploration and classification of major sustainability elements in disaster management education based on a disaster management cycle for attaining disaster resilience among civil engineering students

(Continued)

Table 2. Continued.

No	Author/s and year of publication	Discipline	Level of education	Country/ location of study	Focus of article
41.	J. Taylor et al., 2021	General	Higher education	Finland	Looks at sustainable urban development from the perspective of the skills and competencies required and identifies effective pedagogic practices that could help educate future professionals
42.	L. A. Wolff & Ehrström, 2020	Social sciences	Higher education	Finland, Germany	Aims to create a definition of social sustainability based on a comprehensive literature study, and to discuss the implementation of the concept in higher education settings at theoretical and practical levels

Table 3. Words associated with social sustainability.

Words associated with social sustainability	In no. of articles
'Equality'	21
'Justice'	20
'Inclusion'	18
'Health'	16
'Wellbeing'	13
'Diversity'	12
'Quality of life,' 'equity,' 'equal opportunities'	11
'Equal access,' 'participation,' 'poverty,' 'democracy/democratic rights'	10
'Citizenship'	9
'Cohesion'	7
'Skills,' 'belonging,' 'safety,' 'responsibility,' 'inequalities,' 'security,' 'welfare,' 'values,' 'stakeholder involvement,' 'respect,' 'growth,' 'awareness,' 'space for people,' 'capacity building,' 'future generations,' 'satisfaction,' 'equal distribution,' 'civil society,' 'ethics,' 'life-long learning,' 'peace,' 'governance,' 'decision-making,' 'interaction,' 'fair distribution,' 'quality education,' 'collaboration'	4–6
'Capabilities,' 'recognition,' 'flexibility,' 'design,' 'infrastructure,' 'stable societies,' 'migration,' 'agency,' 'human rights,' 'identity,' 'exclusion,' 'social capital,' 'infrastructure,' 'transportation,' 'human needs,' 'commitment,' 'corporate social responsibility,' 'cohabitation,' 'happiness,' 'recreation,' 'play,' 'employability,' 'abilities,' 'better world,' 'solidarity,' 'mobility,' 'social world,' 'amenities,' 'social and cultural life,' 'autonomy,' 'housing,' 'cooperation,' 'good living conditions,' 'freedom,' 'labour practices,' 'partnership,' 'decent life'	2–3

Table 4. Education disciplines.

Education discipline	Articles no.	No. of articles
Social sciences	2, 3, 4, 5, 6, 11, 13, 18, 19, 25, 29, 30, 32, 33, 37, 39, 42	17
Natural and applied sciences	1, 7, 8, 9, 10, 16, 17, 22, 23, 24, 27, 31, 34, 35, 38, 40	16
Interdisciplinary/unspecified	12, 14, 21, 26, 41	5
Business	20, 28, 36	3
Humanities	15	1

Table 5. Levels of education.

Level of education	Article no.	No. of articles
Early age/preschool	2, 4, 5, 6, 11, 19, 32, 33	8
Elementary	3, 13	2
Higher education	1, 7, 8, 10, 12, 14, 16, 20, 21, 22, 23, 24, 25, 26, 27, 28, 30, 34, 35, 36, 38, 39, 40, 41, 42	25
Lifelong learning	18	1
General	5, 9, 15, 17, 29, 31, 33, 37	8

Table 6. Geographical distribution.

Geographic area	Article no.	No. of articles
Europe	2, 3, 4, 6, 7, 10, 12, 13, 20, 23, 27, 28, 29, 30, 32, 36, 41, 42	18
North America	11, 19, 38	3
South America	35	1
Asia	1, 8, 15, 16, 18, 21, 22, 24, 26, 34, 40	11
Africa	39	1
Oceania	0	0
Unspecified	5, 9, 14, 17, 25, 31, 33, 37	8

Regarding the geographical distribution, studies in European countries dominate numerically with 18 articles and South American and African countries are represented by one article each (Table 6).

The results of the screening of the 42 articles in this review for definitions and interpretations of social sustainability in education can be organised under the categories 'Equality and Opportunities,' 'Social Justice,' and 'Citizenship and Participation.'

Equality and opportunities

The concept of 'equality' is mentioned as a defining feature of social sustainability in 21 articles, and in ten of these articles, equality is mentioned specifically in relation to 'gender equality' (Balakrishnan et al., 2021; Chowdhury et al., 2020; Heldal et al., 2021; Kordi et al., 2022; Liwång, 2020; Lo Presti et al., 2024; Luchoro-Parrilla et al., 2021; Marcelino et al., 2019; Siller et al., 2022; Wolff & Ehrström, 2020).

In other studies, references are made to 'social sustainability' and 'equality' in terms of 'equal opportunities,' but to different resources, such as the opportunity for learners to 'influence their own lives and the communities in which they live' (Boldermo, 2019, p. 3), and to 'contribute to society as a whole' (Kazamia & Kafaridou, 2019, p. 401), thus expanding social sustainability issues in education to their impact on societies. Equality in education can also be related to material issues, such as providing 'better and more job opportunities' (Erjavec, 2020, p. 253), and 'work, basic services, business activities' (Lo Presti et al., 2024, p. 17), or simply put 'equal access to resources and opportunities' for all (Malandrakis et al., 2019, p. 1527).

Some studies relate equality in education to 'lifelong learning opportunities for all' (Harizan & Shah, 2022, p. 117; Simmonds & Ajani, 2022, p. 145; Wolff & Ehrström, 2020, p. 2), and that social sustainability in education is about creating equal 'opportunities for learning and self-development, security, health, social cohesion, cultural diversity, respect to traditions, sense of belonging and social recognition' (Marcelino et al., 2019, p. 2) and about 'teaching individuals to identify injustices (and) inequalities' (Borg & Gericke, 2021, p. 2).

Social justice

Social sustainability in education is also positively related to 'social justice,' which is mentioned in 20 of the articles, in the contexts of providing access to education to achieve social sustainability and social justice, and of creating a tool for learners in creating social sustainability and social justice in society in the future working life (Bergan et al., 2021; Boldermo, 2019; Boldermo & Ødegaard, 2019; Borg & Gericke, 2021; Ceylan & Soygeniş, 2019; Cuenca-Soto et al., 2023; Domingues, 2021; Franck & Osbeck, 2018; Guo et al., 2020; Harizan & Shah, 2022; Larkins et al., 2018; Lo Presti et al., 2024; Malandrakis et al., 2019; Marcelino et al., 2019; Oropilla & Ødegaard, 2021; Pasha & Sreekanth, 2022; Rey-Garcia & Mato-Santiso, 2023; Sadownik & Gabi, 2021; Siller et al., 2022; Wolff & Ehrström, 2020).

Citizenship and participation

'Citizenship' is related to social sustainability in 9 of the articles, and 'participation' in 10 articles. Important for social sustainability issues in education is to 'develop responsible attitudes towards sustainable citizenship' (Biasutti et al., 2019, p. 2), to 'play a role in promoting global citizenship' (Chowdhury et al., 2020, p. 1), and that the pedagogy 'should embody socially critical views of pedagogy that lead to full and active citizenship (linking) memory, agency and reason to the construction of a democratic public sphere' (Larkins et al., 2018, p. 1059). According to Boldermo (2019, p. 2) '(s)ubstantive aspects such as social cohesion, inclusion, belonging and identity are central in defining social sustainability' so that 'all members of society (...) are permitted to participate in the society as peers.'

In some studies, social sustainability in education is focusing on the rights of children to participate (Borg & Gericke, 2021, p. 1; Heldal et al., 2021, p. 2), and for them to acquire 'personally responsible, participatory decision-making skills' (Chowdhury et al., 2020, p. 1).

For others, 'active participation of stakeholders' is highlighted as crucial for social sustainability in education for future work life (Lo Presti et al., 2024, p. 17), and 'social welfare (in projects) is achieved through the procedural aspects of social sustainability, which involves broad participation in decision-making and regulation of scientific and technological activities' (Marcelino et al., 2019, p. 2).

Social sustainability in education is also related to 'finding new ways of living together, strengthening social capital and participation as well as social justice and equity' (Oropilla & Ødegaard, 2021, p. 2). Finally, in one study it is argued that '(i)ndividual liberty and recognition of human potential, democratic participation, equality, good citizenship, and service to others, as well as the growth of knowledge and the resources to allow it, are all prerequisites for social sustainability' (Pasha & Sreekanth, 2022, p. 11460).

Research question 2: What opportunities and challenges for social sustainability in education can be identified? How is social sustainability related to ESD and ESG in education?

The 42 publications in this review were also screened to identify challenges and opportunities for social sustainability in education, and how sustainability can be achieved through education. These can be organised into 'Conceptual Complexity,' 'Measuring Progress,' 'A Lack of Attention to Social Sustainability,' 'Integration with Early Childhood and Primary Education,' 'Social Sustainability in Natural and Applied Science Education,' and 'Critical Transformative Approaches.'

Conceptual complexity

Students sometimes experience difficulties with comprehending and conceptualising social sustainability, finding it challenging to provide a precise meaning of social sustainability, as it is unclear and too complex a concept to elucidate and grasp (Ceylan & Soygeniş, 2019; Chowdhury et al., 2020; Kordi et al., 2022; Malandrakis et al., 2019). Furthermore, it is associated with many different issues (Lo Presti et al., 2024) and often have long-term implications and cover non-linear behaviour, cutting across economic, social, and environmental domains of local to global and often concern urgent matters with high risks (Rey-Garcia & Mato-Santiso, 2020).

Measuring progress

Measuring the levels of social capital and progress is difficult as the most relevant factors to explore are tacit and relational and reflect individual attitudes, perceptions, and behaviours that at best may provide proxy measures (Rey-Garcia & Mato-Santiso, 2020). Furthermore, the social aspect of sustainability does not promise tangible consequences (Ceylan & Soygeniş, 2019), lacks a set series of indicators to measure progress, and therefore has been neglected due to its impracticality (Lo Presti et al., 2024).

In the absence of social indicators, social sustainability issues have been left to market-based strategies to promote social justice, resulting in a failure to pay attention to the relationships, synergies, and trade-offs between the three pillars (Marcelino et al., 2019). Also, the sets of social sustainability indicators that have been proposed for education are not founded in theory, as a clear theoretical framework for social sustainability is missing, but rather in practical understanding of plausibility and current political agendas (Malandrakis et al., 2019; Wolff & Ehrström, 2020).

To better integrate the three pillars of sustainability, it is suggested that a theoretical toolkit should be developed that enables connections between and across the pillars using the concept of 'belonging' (Sadownik & Gabi, 2021). Implementing social sustainability at the course level is not enough, and the only way to reshape and rethink the role of educational institutions is through policy and by including social sustainability in planning and implementation, with ethics at the core (Wolff & Ehrström, 2020).

A lack of attention to social sustainability

One of the major challenges is that too little attention is paid to social sustainability in education compared to environmental and economic sustainability issues (Balakrishnan et al., 2021; Boldermo, 2019; Malandrakis et al., 2019; Marcelino et al., 2019; Syed Zakaria et al., 2020; Wolff & Ehrström, 2020).

Furthermore, it is argued that the social dimension was integrated too late into debates on sustainability, leading to a lack of theoretical and empirical studies in comparison with the other dimensions (Rey-Garcia & Mato-Santiso, 2020). As a result, social sustainability is often considered the least developed dimension of sustainability (Ibrahim, 2020), and simply lacks theoretical justification (Sadownik & Gabi, 2021). Consequently, social sustainability has therefore largely been ignored in education discourses,

practices, and evaluation, leading to a reductionist interpretation of sustainability in education (Franck & Osbeck, 2018; Larkins et al., 2018; Marcelino et al., 2019)

However, it is also suggested that the education for sustainability (ESD) approach has the opportunity and a critical role in providing transdisciplinary skills, with universities having an important role as drivers of sustainable change (Lo Presti et al., 2024; Taylor et al., 2021; Wolff & Ehrstrom, 2020). The establishment of higher education institutions may also contribute to regional sustainable development (Guo et al., 2020; Hung, 2023; Liu et al., 2022) and distance and on-line learning can contribute to social sustainability and opportunities for lifelong learning also in remote locations (Harizan & Shah, 2022; Liu et al., 2022).

Using evidence-based literature and simulation-based learning (Hallinger et al., 2020) and forming real-world learning partnerships with organisations in the community, are also important for students to enable a broader social sustainability impact in education (Chowdhury et al., 2020; Kazamia & Kafaridou, 2019; Rey-Garcia & Mato-Santiso, 2020; Syed Zakaria et al., 2020; Taylor et al., 2021).

Integration with early childhood and primary education

Useful practice-oriented pedagogical approaches must also be developed for early childhood education for sustainability (ECEfs) (Bergan et al., 2021; Boldermo & Ødegaard, 2019; Borg & Gericke, 2021; Ødegaard, 2021), for instance the use of outdoor environments through community projects, problem-based learning, and children's agency and participation (Bergan et al., 2021). Some practical challenges that have been identified including concerns among preschool educators over the extra work that may be required for the integration of education for sustainability in teaching activities, and the general lack of teaching materials about social sustainability for preschool education (Borg & Gericke, 2021).

Providing quality education for all is also challenged by the special needs of children from minorities, pupils with disabilities, and children with sociocultural disadvantage who are at risk of underachieving at school and whose needs pose its own set of challenges (Biasutti et al., 2019). For early childhood education for sustainability, there is also a general lack of research on diversity, multicultural perspectives, and migrant children's situations (Boldermo & Ødegaard, 2019; Domingues, 2021; Haldal et al., 2021).

Forming a community support group among educators for professional formation is suggested as an opportunity to reconceptualize early childhood education for 'Just Sustainabilities' (Domingues, 2021). For the education of migrant children, creating the opportunity for educators to discuss different pedagogical approaches, teaching strategies and practices in intercultural education is an important part of fulfilling social sustainability goals in education (Biasutti et al., 2019; Haldal et al., 2021). The recognition of collaborative exploration of a future-oriented pedagogy could be a driver of social sustainability and can be the best investment any country can to in the name of sustainability (Ødegaard, 2021).

Focusing on migrant children's experiences of belonging is important to create a social sustainable society for all (Boldermo, 2019), and storytelling is brought forward as a method (Malandrakis et al., 2019) to inspire young people to reflect upon what a good society and a good world might be (Franck & Osbeck, 2018). Also, intergenerational interaction between preschool children and older adults provides an opportunity to overcome age separation and segregation, increase life satisfaction, and contribute to social sustainability (Oropilla & Ødegaard, 2021). Happiness and related constructs are also significant predictors of academic performance and social sustainability, having direct implications to SDG 4 and its goal to provide quality education for all (Gamage et al., 2022).

Social sustainability in natural and applied science education

Sustainability approaches in engineering education have been largely limited to technical (Breznik et al., 2021; Liwång, 2020) or economic measures (Kordi et al., 2022). The education system in engineering has often overlooked the dimension of social sustainability, and it is seldom specified as a learning objective in course syllabi (Kordi et al., 2022; Liwång, 2020; Marcelino et al., 2019).

Social topics may also be outside technical faculty comfort zones and require different instructor skills to deal with subjective and normative issues (Siller et al., 2022). In the engineering sector, problems with a lack of manuals and other education material for learners are also offered explanations for the lack of social sustainability issues in curricula (Kordi et al., 2022). It is also argued that where there is a lack of education material (Larkins et al., 2018; Malandrakis et al., 2019), there is a risk of leaving the teaching

to the will and preferences of individual educators, who may or may not be interested in social sustainability issues (Luchoro-Parrilla et al., 2021)

Likewise, in architecture and design education, social aspects of sustainability are given little importance (Ceylan & Soygeniş, 2019; Ibrahim, 2020; (Kazamia & Kafaridou, 2019). Within green chemistry education, there is also no clear relationship with social sustainability, and chemistry has only quite recently rediscovered its social responsibility (Marcelino et al., 2019).

In the engineering sector, social sustainability issues may also face the challenge of being deemed as irrelevant to other subjects (Kordi et al., 2022; Liwång, 2020). To achieve social sustainability, engineering students must understand that issues cannot always be assessed objectively but must be seen contextually and as dependent on social norms, but the value-laden nature of social sustainability is also what presents a challenge to the learners (Liwång, 2020; Rey-Garcia & Mato-Santiso, 2020). However, it is argued that before graduating and joining the industry, students in engineering must take the opportunity to acquire a basic grasp of sustainability to achieve social sustainability criteria (Pasha & Sreekanth, 2022) and to enhance the awareness of engineers' responsibility towards society (Balakrishnan et al., 2021).

For students of architecture the concept of sustainability must be introduced to the students systematically and intensely with theoretical as well as practical knowledge in different stages of architectural education (Ceylan & Soygeniş, 2019). Design curricula should integrate the concept of sustainability by creating cross-disciplinary environments and include stakeholders and real-world systems (Kazamia & Kafaridou, 2019), for instance by designing spaces that reflect the local cultural heritage or local natural environment to create social sustainability and well-being for citizens (Ibrahim, 2020).

Engineering students are also encouraged to think critically about developing cleaner production concepts, job creation, ergonomics, and the quality of employees, in the hope that in the future, engineers have a critical vision regarding sustainability and understand that a good productive system must provide profit without social issues (Rampasso et al., 2019).

What is referred to as 'JEDI' (i.e. Justice, Equality, Democracy, and Inclusion) issues should bring critical components that are important for the development of strong social sustainability in engineering as well as in the larger system of society (Siller et al., 2022), and study visits and role play are also opportunities for learners to integrate social sustainability into the engineering curriculum (Kordi et al., 2022). This does not imply that engineers must be experts in topics such as social justice, but they must be able to collaborate with JEDI experts. This is because engineers without JEDI training may reduce social justice into trivial or formulaic design changes that do not centre human dignity and fail to meet the needs and opportunities of a specific context (Siller et al., 2022). By reflecting on change and the interactions between social and technical system components (Liwång, 2020), engineers can provide solutions for people, since they understand what is (or could be) technically possible, whereas people without engineering training might have to fall back on well-known existing technologies or may suggest ideas that are not viable (Siller et al., 2022). Engineers can also help curb enthusiasm and technological optimism by redefining what is possible and what must be compromised to create sustainability (Marcelino et al., 2019).

Critical transformative approaches

A lack of research about critical transformative approaches, such as ecopedagogy is standing in the way for a necessary change of paradigm in higher education, to increase the awareness of and sensitivity among learners towards sustainability issues (Cuenca-Soto et al., 2023). For example, the use of equity-based critical pedagogy and a gender-fluid persona doll can be used to introduce transgender issues among preschool educators, co-develop equity actions and address social injustice issues with input from children (Domingues, 2021).

Transformative pedagogical processes can also provide opportunities to reform education content and create a decolonised curriculum contextualised in the global South, and to introduce alternative teaching approaches, including storytelling, in university curricula dominated by Eurocentric and Western paradigms (Simmonds & Ajani, 2022). However, the self-transformation of higher education institutions requires a deeper understanding of the social dimensions of sustainability to fully understand its transformative capacities (Rey-Garcia & Mato-Santiso, 2020; Wolff & Ehrström, 2020).

Higher education has been criticised for its inability to provide inclusive, equitable and sustainable quality education, and it is suggested that since happiness is 'the ultimate goal for human development,'

the introduction of a 'sustainability-happiness' index on campus and beyond can provide an opportunity to inform decisions to achieve sustainability (Gamage et al., 2022).

In a study of sustainability textbooks in English, it was argued that we should use a more multi-voiced approach to authorship, an updated integration of the role of culture as a change agent, and the incorporation of the politics of difference to guide the reader more fully to the engagement in social sustainability issues (Larkins et al., 2018). Storytelling within the framework of Education for Sustainable Development (ESD) can also be used as inspiration for young people to envision a good, sustainable society, and how to get there (Franck & Osbeck, 2018; Malandrakis et al., 2019).

Within the field of Early Childhood Education and Care (ECEC), it is argued that collaborative exploration should be integrated with the child-centred early years pedagogy for children to become members of sustainable education practices (Ødegaard, 2021). Intergenerational engagements and programs in kindergartens, exploring intergenerational dialogues, collaborations, and shared knowledge, can also contribute to relational and socially sustainable futures (Oropilla & Ødegaard, 2021). The legacy and integration of culture education curricula also provide opportunities to enhance local identities and traditions, meeting contemporary challenges such as democratic coexistence and social sustainability (Luchoro-Parrilla et al., 2021).

For business students, it is necessary to develop skills in critical thinking about sustainability to meet corporate demands (Rampasso et al., 2019) and the integration of economic, environmental as well as social factors aimed at advancing enterprise performance will lead to the opportunity of achieving 'tri-profitability' goals, moving beyond sustainability towards flourishing businesses and not only 'do well and do (some) good,' but 'do good **to** do well' (Hoveskog et al., 2018, emphasis original).

It is argued that overall, social sustainability must take the opportunity to become a deep ethical element in education, instead of focusing only on specific subjects and special areas (Wolff & Ehrström, 2020) and the role of education is emphasised as an essential inequality preventive mechanism, highlighting the importance of investing in intangibles, primarily human capital (Erjavec, 2020). Education can also play a direct and indirect role in enabling learners to become agents of change for healthier societies and integrating SDGs into the curriculum can therefore be seen as an ethical response to the call of global citizenship (Gürgeç Şimşek & Erkin, 2022). Restorative learning can also help foster a decolonised curriculum that is attuned to sustainable teacher education, contributing to a (re)configuration of sustainable development through aspects such as intercultural understanding, appreciation for diversity, and respect for difference (Simmonds & Ajani, 2022).

Discussion

The research in this review broadly agrees that education is a key driver of social sustainability, which, in turn, is essential for achieving overall sustainability. Recognizing the critical role of education in fostering social sustainability, this scoping review aimed to provide a comprehensive overview of current research on the topic. The findings reveal a wide range of definitions and approaches, highlighting the complexity and diversity of perspectives in the field.

One approach highlighted in the research involves teaching the Sustainable Development Goals (SDGs) by equipping learners with the knowledge and skills needed to promote sustainable development. This includes course content on, for example, human rights, gender equality, peace, global citizenship, and cultural diversity. Another approach focuses on embedding social sustainability principles within the teaching process itself, ensuring that educators develop competencies such as cognitive skills and ethical values. This enables both educators and learners to engage in critical reflection on social sustainability issues. However, determining which knowledge and skills should be taught remains a challenge due to the diverse and complex definitions and interpretations of social sustainability.

This research identifies over 80 distinct concepts that, in various contexts and combinations, define and interpret social sustainability. This diversity underscores the wide range of approaches to integrating social sustainability in education and confirms previous findings that no universally accepted definition exists. Depending on perspective, this variety can either be seen as a wealth of opportunities for incorporating social sustainability into education or as a fragmented landscape that resists standardization and assessment. Nonetheless, the research highlights that the concept's ambiguity—coupled with the

challenges of quantifying, measuring, and assessing social sustainability—contributes to its limited attention in both education and education research.

The various suggestions in this review for integrating social sustainability into education appear to take two distinct directions. One aligns with the ESDG approach, advocating for a more standardized understanding of social sustainability and how it contributes to achieving the SDGs. This perspective emphasizes a universal framework for sustainable development. The other reflects the broader ESD approach, which views social sustainability as context-dependent and adaptable, acknowledging that its meaning and implementation may vary based on specific educational and societal settings.

To address the issue of the ambiguity of the social sustainability concept, researchers in this review suggest developing theoretical and methodological tools—such as manuals and handbooks—to support the assessment of social sustainability, particularly in science education. Aligned with the Education for Sustainable Development Goals (ESDG) approach, these recommendations emphasize the need to standardize social sustainability topics in education and establish clear definitions. This would enable more effective assessment of progress and help achieve the targets set by the Sustainable Development Goals (SDGs).

Some articles, particularly those from social science research and aligned with the broader Education for Sustainable Development (ESD) approach, argue that rather than seeking to standardize definitions, the focus should be on providing concrete examples of how social sustainability can be integrated and applied in specific educational contexts to enhance its relevance. Additionally, science and business articles in this review emphasize the need to strengthen collaboration with communities, engage stakeholders, and incorporate simulations and real-life experiences. These strategies are seen as crucial for fostering a deeper understanding of social sustainability and highlighting its significance in future professional life.

The ESD approach is more characteristic of the social science articles in this review and emphasize the need for new and more critical pedagogical approaches to address social sustainability in education. Recommendations include fostering better collaboration and co-learning among educators, along with awareness-raising initiatives to enhance their engagement with these issues. Some studies suggest that incorporating critical feminist and decolonial perspectives can help educators critically analyse educational content, evaluate curricula, and make necessary revisions. To foster social sustainability in education, proposed critical pedagogical methods include using, for example, persona dolls, storytelling, traditional games, foraging and gardening, collaborative exploration, and dialogue. Additionally, the social science articles address the social sustainability challenges encountered by learners and participants in specific educational settings—such as education for migrant children and young patients—emphasizing the need to promote equality, inclusion, and a strong sense of belonging and citizenship also for marginal groups of learners.

The research in this scoping review varies across disciplines, education levels, study locations, and aspects of social sustainability. The articles also focus on different kinds of participants and relationships within education, including interactions between educators and learners, among educators, between learners, and between learners and external stakeholders—all of which are seen as crucial for achieving social sustainability through education. The social sustainability issues explored in this review encompass both cognitive processes at the individual level, in line with the Inner Development Goals, and social dynamics within educational settings, as well as the knowledge required to prepare learners for their future professions and their role in advancing sustainability in society.

The inclusion criteria for this review were intentionally broad to capture current trends and patterns, and as the results demonstrate, social sustainability issues in education and educational research encompass a wide range of definitions, approaches, and proposed solutions for achieving sustainability. This diversity reflects the richness of the research field, highlighting numerous potential solutions and pathways to advance the SDGs. However, it also presents a challenge for assessing progress, as educators, learners, and educational institutions—while all working towards social sustainability—are addressing different issues, utilizing diverse tools, and pursuing varied goals.

The issues related to conceptual vagueness, along with the calls for standardizing social sustainability indicators and assessment tools, reflect global efforts—led by UNESCO—to universalize education for sustainable development and measure progress. From this perspective, the failure to achieve social sustainability is often viewed as a conceptual challenge in the articles, as seen in the numerous suggestions for improving the integration of social sustainability issues in education by agreeing on its definition and scope.

Overall, while there is criticism regarding the current state of social sustainability in education, the primary focus in the research in this review is on identifying effective ways to integrate and implement social sustainability issues to support the fulfilment of the SDGs, rather than questioning their legitimacy and effectiveness. While previous research has heavily criticized the foundations of ‘sustainable development’ and highlighted the failure of education to achieve social sustainability—and, by extension, overall sustainability—the research in this review focuses not on whether education for social sustainability can or should meet SDG targets and contribute to sustainability, but on how it can do so.

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Author contributions

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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